Jeeho Shin

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EDUCATION

Columbia University New York, NY
MS in Computer Science (Machine Learning Track) *Sep 2026 – Dec 2027*
Courses : High Performance Machine Learning, Deep Learning, Natural Language Processing

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, KR
BS in Computer Science *Aug 2026*
Courses: Machine Learning, Computer Vision, Information Retrieval, Database, Data Structures, Algorithms

EXPERIENCE

AI Research Intern Seongnam, KR
NC AI (AI Research Organization of NCSOFT) – Chatbot Service Team *Sep 2025 – Feb 2026*

- Designed evaluation system for LangGraph-based RAG chatbot to gauge response quality without relying on external LLM API calls, mitigating data privacy risks by fine-tuning open-source LLM using GRPO
- Implemented context-aware decoding pipeline with Transformers library to generate synthetic responses with varying contextual relevance, strengthening model's response-ranking capability via custom reward functions
- Validated fine-tuned model on 500 internal chat samples, achieving 86.3% agreement with human evaluators and 8.8% improvement over standard RAGAS metrics, while cutting evaluation latency by 20%
- Deployed fine-tuned model as inference API with Docker and integrated evaluation module as LangGraph node, enabling real-time, privacy-preserving response quality checks through Langfuse within live chatbot workflow
- Collaborated with 4 engineers, identifying need for real-time visibility into response failures, then built alerting API using Microsoft Teams' Adaptive Cards feature and documented implementation for team-wide use

Undergraduate Researcher Seoul, KR
KAIST AI – Data Mining Lab *Jan 2025 – Aug 2025*

- Designed multimodal recipe recommender system with PyTorch, addressing underutilization of user reviews by integrating review-derived dietary preference signals into user and recipe representation learning
- Constructed preprocessing pipeline for large-scale datasets (~76K users, ~75K recipes), leveraging LLM/VLM with distributed inference across multiple GPUs to summarize user preferences and multimodal recipe features
- Encoded multimodal information summaries into embeddings, then applied graph propagation and cross-view contrastive learning to align multimodal representations with interaction-derived user and recipe embeddings
- Achieved 7–15% Recall@10 gains over 3 advanced multimodal recipe recommendation baselines, reducing training time and peak VRAM usage for learning user and recipe embeddings to one-third of baseline levels

PROJECT

LionPlanner (Link) *Featured Project from Pathfinders Challenge 2026*

- Built full-stack course-planning web app end-to-end, combining React/TypeScript frontend, FastAPI backend with REST APIs, LangGraph-based agentic chatbot, and Supabase vector database, deployed on Vercel and Fly.io
- Scraped syllabi and reviews for 36 Columbia CS courses and created workload-grading pipeline across 5 dimensions using LLM, grounding each score in verbatim evidence with anchor calibration
- Engineered agentic workflow generating responses by calling tools for semantic search, degree-requirement lookup, and schedule-conflict checking to deliver context-aware course recommendations
- Evaluated recommendation quality on 100 queries with LLM evaluator (avg. 4.35/5), tracing each tool's output step by step through LangSmith to pinpoint where failures most often occurred
- Set up GitHub Actions CI pipeline that runs backend and frontend tests with pytest and Vitest on every push

PUBLICATIONS

- From Raw Features to Effective Embeddings: A Three-Stage Approach for Multimodal Recipe Recommendation
  Jeeho Shin**, Kyungho Kim, and Kijung Shin, SIGIR 2026, **Link
- Large language models are zero-shot point-of-interest recommenders
  *Joeun Kim, Youngjin Seo, Yeonsoo Kim, Junhyeok Kang, **Jeeho Shin**, Patara Trirat, and Jae-Gil Lee, DMKD 2025, **Link***

SKILLS

Programming Languages: Python, Java, TypeScript, SQL, C
Machine Learning & AI: PyTorch, Transformers, TRL, LangChain, LangGraph, Langfuse, LangSmith, scikit-learn, W&B
Frameworks & Tools: FastAPI, Supabase, PostgreSQL, React, Docker, Vercel, Fly.io, pytest, Vitest, GitHub Actions, Jira